

IS222:

ASSIGNMENT 01

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1.0 Introduction

1.1 Business Processes

The first step involved in creating a database design is to identify the processes. Database is useful when collecting key data to make the business process successful. These databases are used by businesses to present results and even sometimes it is used to achieve a business goal. The business goal, SuvaTech Solutions want to achieve in this case is to provide a wide range of IT services across various sectors in Fiji.

Each and every business has their own reason for existence, their aims, objectives and the goals they set to achieve in order to accomplish their aims and objectives. According to this scenario, SuvaTech Solutions is to ensure that the company offers IT infrastructure management, software development and AI driven solutions to local businesses ministries and international organizations.

1.2 Role of the Database

- **Manage Data Integrity:** It maintains accurate, consistent and reliable data through the mechanisms like PK/FK.
- **Centralized Data Management:** Stores all the critical data in tables.
- **Facilitating Relationship Between Entities:** all entities are connected to each other ensuring that there is a connection between the entities with its multiplicity.
- **Query-Driven Decision-Making:** Queries are used to draw out answers for critical business questions which supports business operations and reduces manual efforts due to automation.

1.3 Advantages Of Database

- All data is presented in a single database which is easier to access, update and delete.
- Use of queries decreases manual efforts by promoting automation.
- Provides a clear overview of how entities are related to each other via the use of foreign keys.
- Has a summary report of active contracts.
- Have user friendly forms like Client registration and Project Assignment forms.

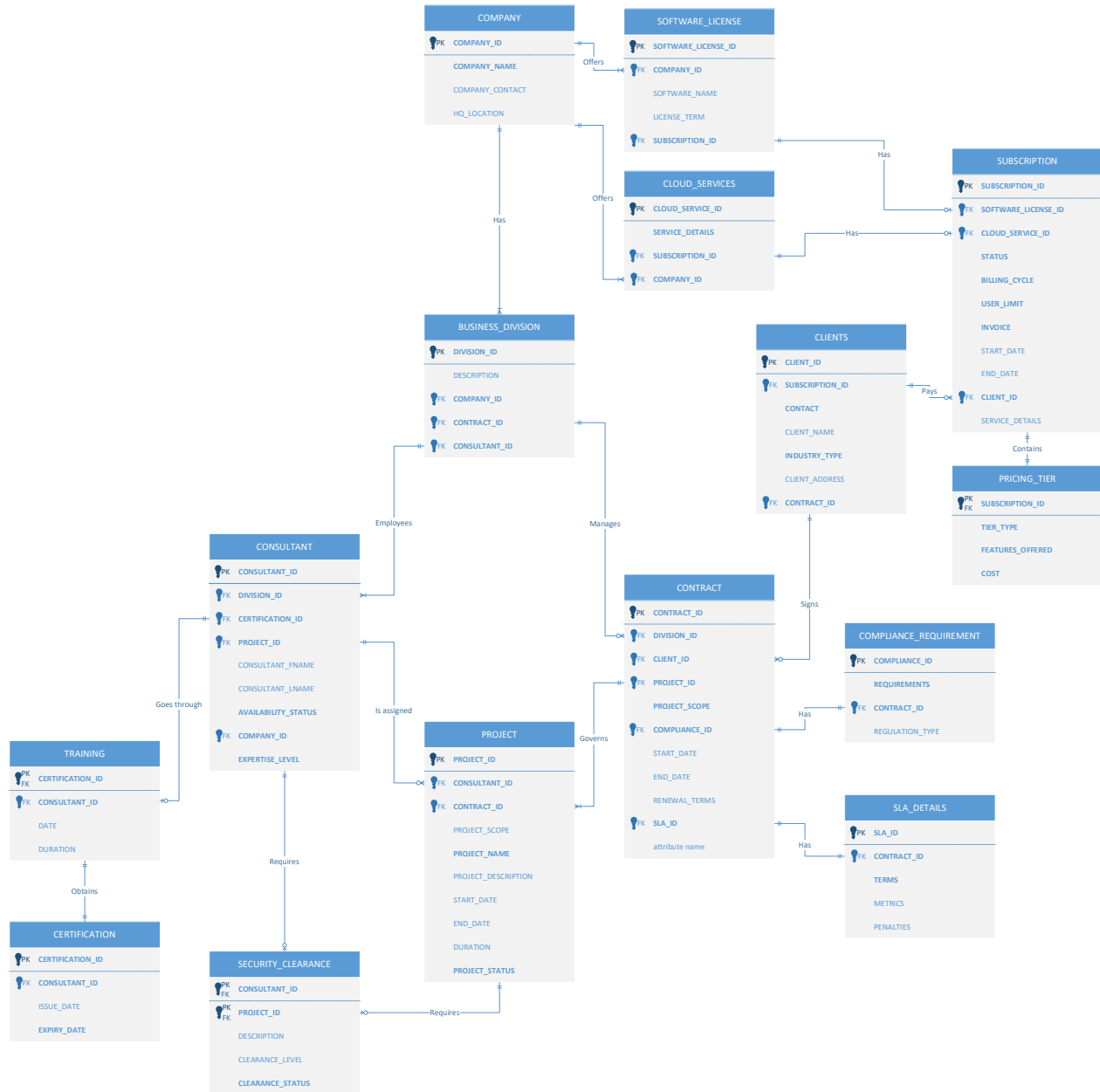
2.0 Business Rules

- Each client can have none to many contracts, while each contract is bound to only one client. *Reason: Clients can sign many contracts.*
- Each contract must be associated with a business division, while one business division can have none to many contracts. *Reason: Each client must be linked to a specific business division.*
- Each consultant can have none to many projects, while one project is associated to none or one consultant. *Reason: Each consultant is assigned and can work on multiple projects.*
- One consultant is associated with one and only one training session. *Reason: It's a requirement for consultants to undergo training sessions*
- One training session is associated with at most one certification, while each certification is associated to one and only one training session. *Reason: Consultants are required to be certified through training session.*
- Each company can have one to many business divisions, while each business division is associated to one and only one company. *Reason: Companies can have multiple business divisions.*
- Each consultant is associated with at most one security clearance. *Reason: Consultants must meet the required security clearance.*
- Companies can offer one to many cloud services, while each cloud service is associated to one company. *Reason: The organization manages cloud services.*
- Companies can offer one to many software licenses, while each software license is associated to one company. *Reason: The organization also manages software licenses.*
- Each client can have none to many subscriptions, while each is associated with one client. *Reason: Clients will have subscription for the services.*

Assumptions:

- Each subscription has one cloud service, while one cloud service can have at most one subscription. *Reason: Cloud services can either have subscription or not.*
- Each subscription is associated with one software license, while one software license can have none to one subscription. *Reason: There might be subscription paid for software license or not.*
- Each project can have none to one security clearance.

3.0 Entity-Relationship Diagram



4.0 Data Dictionary

Entity	Attribute	Content	Type	Format	Range	PK/ FK	Required	Reference table
COMPANY	COMPANY_ID	Company's ID	Short Text	XXX####		PK	Y	
	COMPANY_NAME	Name of the company	Short Text	Xxxxxxxxx			N	
	COMPANY_CONTACT	Company's Contact	Short Text	xxxxxxxxx			Y	
	HQ_LOCATION	Location of the HQ	Short Text	Xxxxxxxxx			N	
BUSINESS_DIVISION	DIVISION_ID	Business Division ID	Short Text	Div#####		PK	Y	
	DESCRIPTION	Description of the Division	Long Text	Xxxxxxxxx			N	
	COMPANY_ID	Company's ID	Short Text	XXX####		FK	Y	COMPANY
	CONTRACT_ID	ID of the Assigned contracts	Short Text	X#####		FK	Y	CONTRACT
CONSULTANT	CONSULTANT_ID	ID of the consultants working	Short Text	C#####		FK	Y	CONSULTANT
	CONSULTANT_ID	Consultant's ID	Short Text	C#####		PK	Y	
	DIVISION_ID	Business Division ID	Short Text	Div#####		FK	Y	BUSINESS_DIVISION
	CERTIFICATION_ID	Certification ID	Short Text	CR#####		FK	Y	CERTIFICATION
PROJECT	PROJECT_ID	ID of the project	Short Text	PR#####		FK	Y	PROJECT
	CONSULTANT_FNAME	First Name	Short Text	Xxxxxxxxx			N	
	CONSULTANT_LNAME	Last Name	Short Text	Xxxxxxxxx			N	
	AVAILABILITY_STATUS	Availability Status	Bool	Yes/No			Y	
COMPANY	COMPANY_ID	Company ID	Short Text	XXX####		FK	Y	COMPANY
	EXPERTISE_LEVEL	Level Of Expertise	Short Text	Lv.Xxxxxxx			Y	
TRAINING	CERTIFICATION_ID	Certification ID	Short Text	CR#####		PK/FK	Y	CERTIFICATION
	CONSULTANT_ID	ID of the Consultant	Short Text	C#####		FK	Y	CONSULTANT
	DATE	Training Date	Date	mm-dd-yyyy			N	
	DURATION	Training Duration	Number	#####	0-9999999		N	
CERTIFICATION	CERTIFICATION_ID	Certification ID	Short Text	CR#####		PK	Y	
	CONSULTANT_ID	ID of the Consultant	Short Text	C#####		FK	Y	CONSULTANT
	ISSUE_DATE	Date of Issue	Date	mm-dd-yyyy			N	
	EXPIRY_DATE	Date of Expiry	Date	mm-dd-yyyy		FK	Y	
SECURITY_CLEARANCE	PROJECT_ID	Project ID	Short Text	Pr#####		PK/FK	Y	PROJECT
	CONSULTANT_ID	ID of the Consultant	Short Text	C#####		PK/FK	Y	CONSULTANT
	DESCRIPTION	Description of the clearance	Long Text	Xxxxxxxxx			N	
	CLEARANCE_LEVEL	Clearance level	Integer	#	1-4		Y	
CLEARANCE_STATUS	CLEARANCE_STATUS	Clearance Status	Bool	Yes/No			Y	
SOFTWARE_LICENSE	SOFTWARE_LICENSE_ID	License ID of the software	Short Text	Li#####		PK	Y	
	SOFTWARE_NAME	Name of the software	Short Text	Xxxxxxxxx			N	
	LICENSE_TERM	Terms related to software	Long Text	Xxxxxxxxx			N	
	SUBSCRIPTION_ID	Subscription ID	Short Text	Sub#####		FK	Y	SUBSCRIPTION
COMPANY_ID	COMPANY_ID	Company ID	Short Text	XXX####		FK	Y	COMPANY
CLOUD_SERVICES	CLOUD_SERVICE_ID	Cloud Service ID	Short Text	Ser#####		PK	Y	
	SERVICE_DETAILS	Details regarding Service	Long Text	Xxxxxxxxx			Y	
	SUBSCRIPTION_ID	Subscription ID	Short Text	Sub#####		FK	Y	SUBSCRIPTION
	COMPANY_ID	Company ID	Short Text	XXX####		FK	Y	COMPANY
SUBSCRIPTION	SUBSCRIPTION_ID	Subscription ID	Short Text	Sub#####		PK	Y	
	SOFTWARE_LICENSE_ID	License ID of the software	Short Text	Li#####		FK	Y	SOFTWARE_LICENSE
	CLOUD_SERVICE_ID	Cloud Service ID	Short Text	Ser#####		FK	Y	CLOUD_SERVICES
	UPGRADE_STATUS	Upgrade Status	Bool	Yes/No			Y	
BILLING_CYCLE	BILLING_CYCLE	Billing Cycle	Short Text	Xxxxxxxxx			Y	
	USER_LIMIT	Number of users	Number	#####			Y	
	INVOICE	Invoice/Receipt	Short Text	XXxxxxxxxx			Y	
	START_DATE	Day of subscription	Date	mm-dd-yyyy			N	
END_DATE	END_DATE	Expiry date of subscription	Date	mm-dd-yyyy			N	
	CLIENT_ID	ID of the client	Short Text	XXX####		FK	Y	CLIENTS
	SERVICE_DETAILS	Details of the services	Long Text	Xxxxxxxxx			N	
PRICING_TIER	SUBSCRIPTION_ID	Subscription ID	Short Text	Sub#####		PK/FK	Y	SUBSCRIPTION
	TIER_TYPE	Type of pack	Short Text	Xxxxxxxxx			Y	
	FEATURES_OFFERED	Offered features	Long Text	Xxxxxxxxx			Y	
	COST	Price	Currency	\$#####.##			Y	
CLIENTS	CLIENT_ID	ID of the client	Short Text	XX#####		PK	Y	
	SUBSCRIPTION_ID	Subscription ID	Short Text	Sub#####		FK	Y	SUBSCRIPTION
	CONTACT	Contact details of client	Short Text	Xxxxxxxxx			Y	
	CLIENT_NAME	Name of the client	Short Text	Xxxxxxxxx			N	
INDUSTRY_TYPE	INDUSTRY_TYPE	Type of client Industry	Short Text	Xxxxxxxxx			Y	
	CLIENT_ADDRESS	Address of the client	Short Text	Xxxxxxxxx			N	
	CONTRACT_ID	ID of the contract associated	Short Text	CO#####		FK	Y	CONTRACT
CONTRACT	CONTRACT_ID	Contract ID	Short Text	CO#####		PK	Y	
	DIVISION_ID	Business Division ID	Short Text	Div#####		FK	Y	BUSINESS_DIVISION
	CLIENT_ID	ID of the client associated	Short Text	XXX####		FK	Y	CLIENTS
	PROJECT_ID	ID of the project managed	Short Text	Pr#####		FK	Y	PROJECT
PROJECT_SCOPE	PROJECT_SCOPE	Scope of the project	Long Text	Xxxxxxxxx			Y	
	COMPLIANCE_ID	Compliance ID	Short Text	CRQ#####		FK	Y	COMPLIANCE_REQUIREMENT
	START_DATE	Contract Starting date	Date	mm-dd-yyyy			N	
	END_DATE	Contract End Date	Date	mm-dd-yyyy			N	
RENEWAL_TERMS	RENEWAL_TERMS	Renewal terms	Short Text	Xxxxxxxxx			N	
	SLA_ID	ID of SLA	Short Text	SLA#####		FK	Y	SLA_DETAILS
PROJECT	PROJECT_ID	Project ID	Short Text	Pr#####		PK	Y	
	CONSULTANT_ID	Linked consultant ID	Short Text	C#####		FK	Y	CONSULTANT
	CONTRACT_ID	Managing contract ID	Short Text	CO#####		FK	Y	CONTRACT
	PROJECT_SCOPE	Project Scope	Long Text	Xxxxxxxxx			N	
PROJECT_NAME	PROJECT_NAME	Name of the Project	Short Text	Xxxxxxxxx			Y	
	PROJECT_DESCRIPTION	Brief description	Long Text	Xxxxxxxxx			N	
	START_DATE	Project starting Date	Date	mm-dd-yyyy			N	
	END_DATE	Project Ending date	Date	mm-dd-yyyy			N	
DURATION	DURATION	Project duration	Number	#####	0-9999999		N	
	PROJECT_STATUS	Project Status	Short Text	Xxxxxxxxx			Y	
COMPLIANCE_REQUIREMENT	COMPLIANCE_ID	Compliance ID	Short Text	CRQ#####		PK	Y	
	REQUIREMENTS	Requirements	Long Text	Xxxxxxxxx			Y	
	CONTRACT_ID	Associated Contract ID	Short Text	CO#####		FK	Y	CONTRACT
	REGULATION_TYPE	Compliance regulation type	Short Text	Xxxxxxxxx			N	
SLA_DETAILS	SLA_ID	SLA ID	Short Text	SLA#####		PK	Y	
	CONTRACT_ID	Associated Contract ID	Short Text	CO#####		FK	Y	CONTRACT
	TERMS	Terms associated	Short Text	Xxxxxxxxx			Y	
	METRICS	SLA metrics	Short Text	Xxxxxxxxx			N	
PENALTIES	PENALTIES	Penalties	Short Text	Xxxxxxxxx			N	

5.0 Conclusion

This is one of these recent projects that I did in which I created a database system for an organization according to the specifications provided to me. I had to refer back to my old lab work material along with utilizing YouTube and Google in order to complete this work. However, the project was a hard nut to crack not so much due to the fact that I had a whole list of other assignments and exams to face, but since the submission date was prolonged, I was able to complete it successfully. Certain issues posed in relation to the topic of the assignment are ones that resulted in the difficulty; otherwise, I was not quite aware of MS Access, a problem which took time to be solved. And lo and behold, I not only could not get to the correct version of Visio that I could download to my own PC but worked hard to get myself Visio 2016 which got me successfully to the ERD creation. The data dictionary was the most time-consuming for me, but I was at last able to determine how to update it so that it could be utilized both in MS Word and Access.